AEO "Nazarbayev Intellectual Schools"

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CENTRE NUMBER							CANDIDATE NUMBER				

6174752788

CHEMISTRY **Grade 12**

Paper 1 **May 2023**

1 hour

Candidates answer on the Question Paper.

Additional Materials: Calculator
Ruler
Data Booklet

12CHEM/01

READ THESE INSTRUCTIONS FIRST

Write your centre number and candidate number in the spaces at the top of the page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

Calculators may be used.

The total number of marks for this paper is 40.

Answer all questions in English.

For Examiner's Use	
Section A	
Section B	
Total	

This document consists of **16** printed pages.

Section A

For Examiner's Use

For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider to be correct by putting a tick (✓) in the correct box.

1 The electronic configuration for chlorine, Cl, is $1s^2 2s^2 2p^6 3s^2 3p^5$.

Which electrons are the shielding electrons?

A $1s^2\,2s^2$

B $3s^2\,3p^5$

C $1s^2\,2s^2\,2p^6$

D $2s^2\,2p^6\,3s^2\,3p^5$

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

2 What is the ionic equation for the reaction between hydrochloric acid and aqueous sodium hydroxide?

A $K^+(aq) + Cl^-(aq) \rightarrow KCl(s)$

B $K^+(aq) + OH^-(aq) \rightarrow KOH(s)$

C $H^+(aq) + Cl^-(aq) \rightarrow HCl(l)$

D $H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

3 Which compound does **not** show E/Z isomerism?

A 1,2-dichloroethene

B 2-methylpropene

C but-2-ene

D hexa-1,3-diene

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

For Examiner's Use

4 What is the definition for oxidation?

A gain of hydrogen

B loss of oxygen

C increase in oxidation number

D gain of electrons

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

5 Which aqueous halide ion produces a precipitate with aqueous silver nitrate that will **not** dissolve in excess concentrated aqueous ammonia?

A F^-

B Cl^-

C Br^-

D I^-

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

6 Calcium carbonate decomposes as shown in the equation.

$$CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$$

$\Delta H^\ominus = +178.0$ kJ/mol $\Delta S^\ominus = +165.0$ J/K × mol

What is the temperature, in K, at which the reaction becomes feasible?

A 1.078

B 107.9

C 1070

D 1079

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

For Examiner's Use

7 Why does $[Fe(H_2O)_6]^{3+}(aq)$ have a lower pH than $[Fe(H_2O)_6]^{2+}(aq)$?

A $[Fe(H_2O)_6]^{3+}(aq)$ is less stable than $[Fe(H_2O)_6]^{2+}(aq)$

B Fe^{3+} has a smaller ionic radius and a lower charge density than Fe^{2+}

C Fe^{3+} is more polarising and has a higher charge density than Fe^{2+}

D Fe^{3+} has a larger ionic radius and a much stronger attraction for electrons in a ligand than Fe^{2+}

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

8 The reversible reaction between sulfur dioxide and oxygen forms an equilibrium mixture in a closed container.

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g) \qquad \Delta H = -98 \text{ kJ mol}^{-1}$$

What happens to the position of equilibrium and the value of K_c as the temperature is increased at constant pressure?

	Position of equilibrium	**Value of K_c**
A	shifts to the left	decreases
B	shifts to the left	increases
C	shifts to the right	decreases
D	shifts to the right	increases

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

9 The table shows some experimental data about the reaction between **A**(aq) and **B**(aq).

What is the total order of the reaction?

Experiment number	**Concentration of A / mol dm^{-3}**	**Concentration of B / mol dm^{-3}**	**Initial rate / mol dm^{-3}c^{-1}**
1	0.02	0.04	4
2	0.02	0.08	8
3	0.04	0.04	8

A 0

B 1

C 2

D 3

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

For Examiner's Use

10 Which combination of solutions forms a buffer solution?

A 100 cm^3 of 0.10 mol dm^{-3} hydrochloric acid with 50 cm^3 of 0.10 mol dm^{-3} sodium hydroxide.

B 100 cm^3 of 0.10 mol dm^{-3} ethanoic acid with 50 cm^3 of 0.10 mol dm^{-3} sodium hydroxide.

C 50 cm^3 of 0.10 mol dm^{-3} hydrochloric acid with 100 cm^3 of 0.10 mol dm^{-3} sodium hydroxide.

D 50 cm^3 of 0.10 mol dm^{-3} ethanoic acid with 100 cm^3 of 0.10 mol dm^{-3} sodium hydroxide.

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

11 Which list shows the elements in order of increasing electronegativity?

A I < Br < C*l* < F

B F < C*l* < Br < I

C F < Br < C*l* < I

D I < C*l* < Br < F

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

12 The first five successive ionization energies for element **X**, in kJ mol^{-1}, are shown.

1st ionization energy	2nd ionization energy	3rd ionization energy	4th ionization energy	5th ionization energy
738	1450	7733	10543	13630

What is the formula of the chloride of **X**?

A $\mathbf{X}Cl$

B $\mathbf{X}Cl_2$

C $\mathbf{X}Cl_3$

D $\mathbf{X}_2Cl_3$

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

For Examiner's Use

13 Which statement is correct for elements in Group 14?

A boiling points of elements increase down the group

B metallic character of element decreases down the group

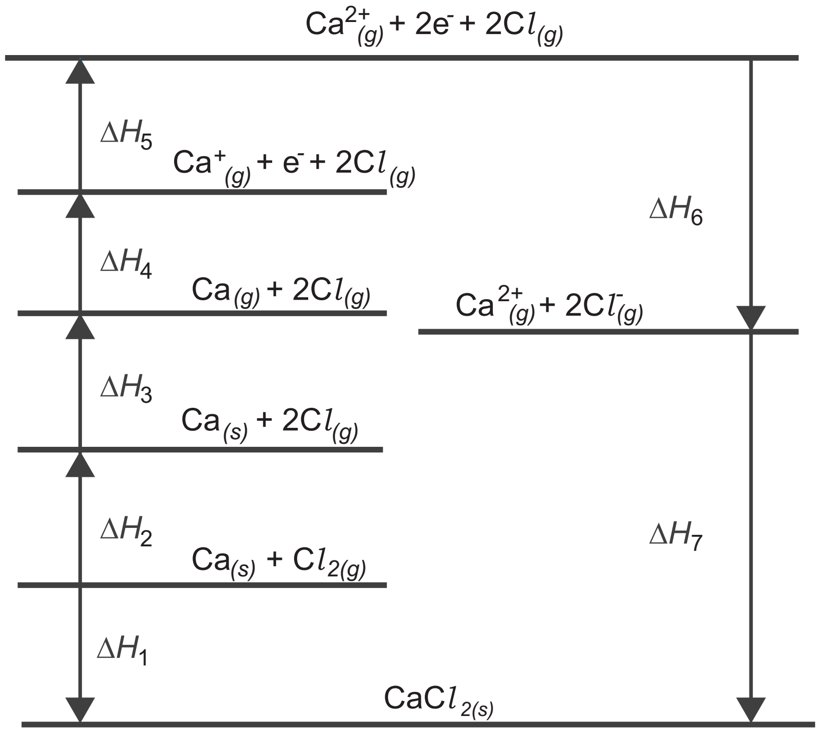
C first ionization energy of elements increases down the group

D stability of the oxide with the element in the +2 oxidation state increases down the group

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

14 The diagram shows a Born-Haber cycle for calcium chloride. It is not drawn to scale.
All units are in kJ mol^{-1}



Which enthalpy change is correctly labelled on the diagram?

A ΔH_4 – bond dissociation enthalpy of chlorine

B ΔH_5 – electron affinity of chlorine

C ΔH_6 – enthalpy of formation of calcium chloride

D ΔH_7 – lattice enthalpy of calcium chloride

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

For Examiner's Use

15 This question is about the reaction of methane with chlorine in sunlight.

$$CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$$

What is the best description of this reaction?

A electrophilic addition.

B electrophilic substitution.

C free radical addition.

D free radical substitution.

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

16 Consider the following Group 2 compounds.

Group 2 hydroxides	**Group 2 sulfates**
$Mg(OH)_2$	$MgSO_4$
$Ca(OH)_2$	$CaSO_4$
$Ba(OH)_2$	$BaSO_4$

Which statement is correct?

A The solubility for both Group 2 hydroxides and Group 2 sulfates increases down the group.

B The solubility for Group 2 hydroxides increases and for Group 2 sulfates decreases down the group.

C The solubility for Group 2 hydroxides decreases and for Group 2 sulfates increases down the group.

D The solubility for both Group 2 hydroxides and Group 2 sulfates decreases down the group.

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

17 Which class of organic compounds are functional group isomers for carboxylic acids?

A ethers

B ketones

C alcohols

D esters

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

For Examiner's Use

18 Use of the data booklet is relevant to this question.

What is the value of the E_{cell} for an electrochemical cell with the half-reactions shown.

$$Mn^{3+}(aq) + e^{-} \rightarrow Mn^{2+}(aq)$$

$$Co^{2+}(aq) + 2e^{-} \rightarrow Co(s)$$

A +1.77 V

B –1.77 V

C +1.21 V

D –1.21 V

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

19 Which compound has the most exothermic standard lattice enthalpy?

A BeO

B NaC*l*

C KBr

D MgS

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

20 Which row in the table is correct?

	Complex	**Shape**	**Co-ordination number**	**Oxidation state of central cation**
A	$[Ag(CN)_2]^-$	Linear	2	–1
B	$[CuCl_4]^{2-}$	Tetrahedral	4	–2
C	$[Cr(C_2O_4)_3]^{3-}$	Octahedral	3	+3
D	$[Cu(NH_3)_4(H_2O)_2]^{2+}$	Octahedral	6	+2

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

21 The Haber process is used to manufacture ammonia from nitrogen and hydrogen. The reaction is reversible and the forward reaction is exothermic. A high pressure and a compromise temperature of 400 to 450 °C is used. The temperature is changed to 600 °C.

What effect does this have on ammonia production?

	Rate of production	**Equilibrium yield**
A	increases	decreases
B	increases	increases
C	decreases	decreases
D	decreases	increases

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

22 Benzene reacts with a mixture of concentrated nitric acid and concentrated sulfuric acid to make nitrobenzene.

Which ion produced in the mixture of concentrated nitric acid and concentrated sulfuric acid attacks the benzene ring?

A nitrate ion

B nitride ion

C nitrite ion

D nitronium ion

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

23 Which statement about the primary structure of protein is correct?

A The structure of proteins which are composed of two or more smaller protein chains.

B The sequence of amino acids is joined together by peptide bonds to form a polypeptide chain.

C Occurs when structures such as alpha-helices and beta-pleated sheets are present in a protein.

D The structure that is held together by interactions between the side chains of amino acids.

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

For Examiner's Use

24 Which statement about buckminsterfullerene is **not** correct?

A Buckminsterfullerene has a strong structure due to its covalent bonds and spherical shape.

B Buckminsterfullerene has weak van der Waals' dispersion forces between its molecules.

C A molecule of buckminsterfullerene is composed of only hexagonal rings of carbon atoms.

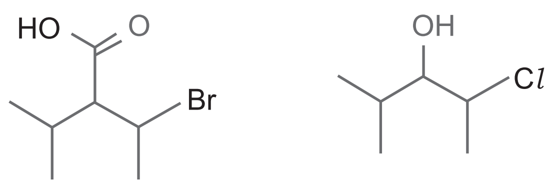
D Buckminsterfullerene is insoluble in water because it cannot form hydrogen bonds with water.

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

25 Which reagent will distinguish **A** from **B**?

A **B**



A Na

B Na_2CO_3(aq)

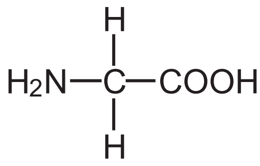
C acidified $AgNO_3$(aq)

D Fehling's solution

A ☐ **B** ☐ **C** ☐ **D** ☐

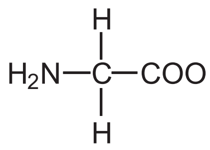
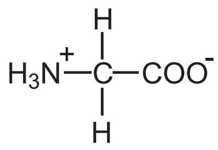
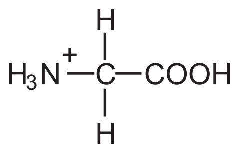
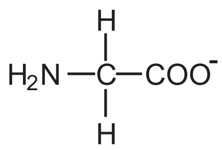
[1]

26 In the figure below given the formula of glycine.



glycine

Which row shows the structure of glycine in an aqueous solution of pH 4?

A	
B	
C	
D	

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

27 Which row about the reducing strength of halide ions is correct?

	Strength as reducing agents	**Explanation**
A	$Cl^- > Br^- > I^-$	I^- ions can reduce Cl^- ions in aqueous solution
B	$Cl^- > Br^- > I^-$	larger anions lose electrons more easily than smaller anions
C	$I^- > Br^- > Cl^-$	I^- ions can reduce Cl^- ions in aqueous solution
D	$I^- > Br^- > Cl^-$	larger anions lose electrons more easily than smaller anions

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

For Examiner's Use

For Examiner's Use

28 The aqueous ethylenediaminetetraacetate ion, $EDTA^{4-}(aq)$, forms a complex ion with $Fe^{2+}(aq)$ with the formula $[Fe(EDTA)]^{2-}(aq)$.

What can be deduced from this information alone?

A $EDTA^{4-}(aq)$ is a monodentate ligand

B $[Fe(EDTA)]^{2-}(aq)$ has a coordination number of 6

C $[Fe(EDTA)]^{2-}(aq)$ has a tetrahedral shape

D $[Fe(EDTA)]^{2-}(aq)$ is less stable than $[Fe(H_2O)_6]^{2+}(aq)$

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

29 Which statement is correct?

A CH_3CH_2CHO is propanone

B $C_2H_5COC_2H_5$ is butanone

C $(CH_3)_2CHCH_2CHO$ is 3-methylbutanal

D $C_2H_5COCH_3$ is butan-3-al

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

30 Lactic acid, $CH_3CH(OH)COOH$, is a weak acid. A 0.375 mol dm^{-3} solution of lactic acid has a pH of 2.14 at 25 °C.

What is the acid dissociation constant, K_a, in mol dm^{-3} at this temperature?

A 1.40×10^{-4}

B 7.24×10^{-4}

C 1.90×10^{-2}

D 3.85

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

For Examiner's Use

Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct).

The responses **A** to **D** should be selected on the basis of

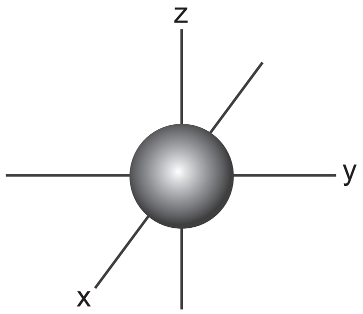
A	B	C	D
1, **2** and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

No other combination of statements is used as a correct response.

Use of the Data Booklet may be appropriate for some questions.

31 The diagram shows the shape of a s-orbital in a ***s*** sub-shell.

Which statement(s) about the ***s*** sub-shell is (are) correct?



1 has a spherical shape

2 holds only two electrons

3 has the lowest energy in every shell

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

32 Magnesium hydroxide is used in some indigestion tablets as an antacid.

Which statement(s) is (are) correct?

1 Magnesium hydroxide is completely dissolved in the oral cavity and delivered to the stomach in a liquid state.

2 Magnesium hydroxide is involved in a neutralization reaction with hydrochloric acid in the stomach.

3 Magnesium hydroxide is less soluble than barium hydroxide.

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

For Examiner's Use

33 Graphite, diamond, and silica have a giant covalent structure.

Which statement(s) is (are) correct?

1 Substances with a giant covalent structure have a very high melting point.

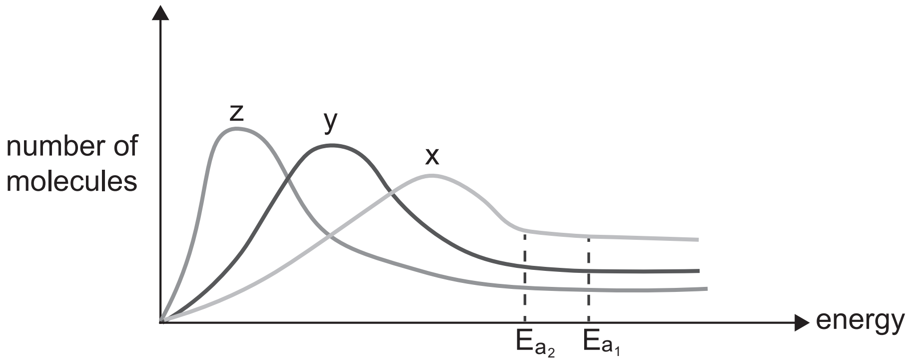
2 Substances with a giant covalent structure conduct electricity due to delocalized electrons.

3 Substances with a giant covalent structure dissolve in polar solvents.

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

34 Which statement(s) about the Maxwell-Boltzmann distribution of one mole of a gaseous reaction is (are) correct?



1 **x** is the distribution with the highest temperature.

2 The number of particles in **z** with enough activation energy is less than in **y**.

3 $\mathbf{E_{A2}}$ represents the presence of a catalyst, whereas $\mathbf{E_{A1}}$ is in the absence of a catalyst.

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

35 Which alcohol(s) is (are) oxidized to a carboxylic acid?

1 $CH_3C(CH_3)_2CH_2OH$

2 $CH_3C(CH_3)(OH)CH_3$

3 $CH_3CH_2CH(OH)CH_3$

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

For Examiner's Use

36 Which reaction(s) show(s) sulfuric acid, H_2SO_4, behaving as an oxidizing agent?

1 $2HBr + H_2SO_4 \rightarrow Br_2 + SO_2 + 2H_2O$

2 $8HI + H_2SO_4 \rightarrow 4I_2 + H_2S + 4H_2O$

3 $2NH_3 + H_2SO_4 \rightarrow (NH_4)_2SO_4$

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

37 Which statement(s) is (are) correct about chirality and the action of drugs?

1 A chiral molecule is non-superimposable on its mirror image.

2 A pair of enantiomers have some properties that are different.

3 Drugs presented in the form of enantiomers may have different effects on the human body.

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

38 Propanone reacts with HCN.

Which statement(s) is (are) correct about the product of this reaction?

1 It is chiral.

2 It contains hydroxyl and nitrile groups.

3 It does not show optical isomerism.

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

39 Compound **G** has a molar mass of 58 g mol^{-1}.

The percentage composition by mass of **G** is carbon 62.07%, hydrogen 10.34% and oxygen 27.59%

G reacts with sodium tetrahydridoborate to give a secondary alcohol.

What can be deduced from this information?

1 **G** is an aldehyde

2 **G** is a carbonyl compound

3 **G** is propanone

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

For Examiner's Use

40 Amino acids contain both an amine group and a carboxylic acid group.

Which statement(s) is (are) correct about amino acids?

1 The carboxylic acid group reacts with an alkali to form a conjugate base.

2 The amino group reacts with an acid to form a salt of the conjugate acid.

3 Amino acids are amphoteric.

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]